

Subject Code	AF5643						
Subject Title	Advanced Topics in Applied Economics						
Credit Value	3						
Level	5						
Normal Duration	One Semester						
Pre-requisite / Co-requisite/ Exclusion	None						
Objectives	This subject contributes to Learning Outcome 2 by developing students' skills in data science and the digital economy. It does so by taking students on a journey from understanding potential outcomes and causal effects to applying advanced methods like regression discontinuity designs, difference-in-differences, and synthetic control methods.						
Intended Learning Outcomes	Upon completion of the subject, you will be able to: a. understand fundamental theories about causal inference. b. apply causal inference related econometrics methods to real datasets. c. understand problems establishing causation often encountered in research d. use software in order to implement the methods learnt from the course.						
Subject Synopsis/ Indicative Syllabus	• Causal inference: Potential Outcomes and Causal Effects; Defining and measuring causal effects; Instrumental Variables. • Difference-in-Differences; • Synthetic Control Methods; • Regression Discontinuity Designs; • Randomized experiments; • Observational Studies and Matching: Propensity score matching and other matching methods • Machine Learning and Causal Inference; • Student presentations.						
Teaching/Learning Methodology	This subject will be taught in both lecture and seminar formats, emphasizing key issues in casual influence and latest development in applied economics.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e

	Attendance and participation	10	✓	✓	✓	✓		
	Presentation and written Report	40	✓	✓	✓	✓		
	Final Exam	50	✓	✓	✓			
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The assessment methods have been carefully chosen to evaluate the intended learning outcomes in a comprehensive manner. Continuous assessment will be employed to gauge your practical abilities, specifically in data handling, programming, and interpretation skills. This method allows for ongoing feedback and the opportunity to improve and refine your skills over time. On the other hand, the final exam is designed to assess your conceptual understanding of the course content. These exams will test your ability to recall, analyze, and apply the theoretical knowledge you have acquired throughout the course. This combination of continuous assessment and examinations ensures a balanced evaluation of both practical application and theoretical understanding, aligning with the intended learning outcomes.							
Student Study Effort Expected	Class contact:							
	▪ Lectures / Seminars					39 Hrs.		
	Other student study effort:							
	▪ Reading of reference materials and Preparation for presentation and written report					72 Hrs.		
	Total student study effort					111 Hrs.		

Reading List and References	Main textbook: Cunningham, S. (2021). <i>Causal inference: The mixtape</i>. Yale university press. Other textbooks: Pearl, J., Glymour, M., & Jewell, N. P. (2016). <i>Causal inference in statistics: A primer</i>. John Wiley & Sons. Brumback, B. A. (2021). <i>Fundamentals of Causal Inference: With R</i>. Chapman and Hall/CRC. Further reading and potential papers to be discussed. Imbens, G. (2014). <i>Instrumental variables: An econometrician's perspective</i> (No. w19983). National Bureau of Economic Research. Angrist, J. D., & Krueger, A. B. (1991). Does compulsory school attendance affect schooling and earnings?. <i>The Quarterly Journal of Economics</i>, 106(4), 979-1014. Callaway, B. (2023). Difference-in-differences for policy evaluation. <i>Handbook of Labor, Human Resources and Population Economics</i>, 1-61. Roth, J., Sant'Anna, P. H., Bilinski, A., & Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. <i>Journal of Econometrics</i>, 235(2), 2218-2244. Braghieri, L., Levy, R. E., & Makarin, A. (2022). Social media and mental health. <i>American Economic Review</i>, 112(11), 3660-3693. Gilchrist, D., Emery, T., Garoupa, N., & Spruk, R. (2023). Synthetic Control Method: A tool for comparative case studies in economic history. <i>Journal of Economic Surveys</i>, 37(2), 409-445. Abadie, A., Diamond, A., & Hainmueller, J. (2015). Comparative politics and the synthetic control method. <i>American Journal of Political Science</i>, 59(2), 495-510.
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